

Produkte
Products

Prüfbericht - Nr.: 50183074 001 <i>Test Report No.:</i>		Seite 1 von 25 <i>Page 1 of 25</i>	
Auftraggeber: <i>Client:</i>		Xinxiang hongli supply source technology co., ltd Dayang Village South, zhaipo Town, Xinxiang, China	
Gegenstand der Prüfung: CYLINDRICAL LITHIUM-ION RECHARGEABLE CELL <i>Test item:</i>			
Bezeichnung: <i>Identification:</i>	IMR18650-1200mAh, IMR18650-1500mAh	Serien-Nr.: <i>Serial No.:</i>	N/A
Wareneingangs-Nr.: <i>Receipt No.:</i>	164141685	Eingangsdatum: <i>Date of receipt:</i>	2018-09-29
Prüfart: <i>Testing location:</i>	Guangzhou MCM Certification & Testing Co., Ltd. 1 F No.13, Zhong San Section, Shi Guang Road, Zhong Cun Street, Panyu District, Guangzhou, Guangdong CHINA		
Prüfgrundlage: <i>Test specification:</i>	UL 1642: 2012 R6.15		
Prüfergebnis: <i>Test Result:</i>	Der Prüfgegenstand entspricht oben genannter Prüfgrundlage(n). <i>The test item passed the test specification(s).</i>		
Prüflaboratorium: <i>Testing Laboratory:</i>	TÜV Rheinland (Shenzhen) Co., Ltd. 1F East & 2-4F, Cybio Technology Building No.1, No.16 Kejibei 2nd Road, High-Tech Industrial Park North Nanshan District, 518057, Shenzhen, China		
geprüft/ tested by:		kontrolliert/ reviewed by:	
 2018-11-08 Caroline Chen/ Project Engineer		 2018-11-08 Daniel Dai / Reviewer	
Datum <i>Date</i>	Name/Stellung <i>Name/Position</i>	Unterschrift <i>Signature</i>	Datum <i>Date</i>
Datum <i>Date</i>	Name/Stellung <i>Name/Position</i>	Unterschrift <i>Signature</i>	Datum <i>Date</i>
Sonstiges/ Other Aspects: TÜV Rheinland TÜVus mark approval procedure. The complete test report includes the following documents: - UL1642 Test report (25 pages); - Attachment 1: Equipment lists (2 pages); - Attachment 2: Photo documentation (2 pages).			
Abkürzungen:		Abbreviations:	
P(ass) = entspricht Prüfgrundlage	F(ail) = entspricht nicht Prüfgrundlage	P(ass) = passed	F(ail) = failed
N/A = nicht anwendbar	N/T = nicht getestet	N/A = not applicable	N/T = not tested
Dieser Prüfbericht bezieht sich nur auf das o.g. Prüfmuster und darf ohne Genehmigung der Prüfstelle nicht auszugsweise vervielfältigt werden. Dieser Bericht berechtigt nicht zur Verwendung eines Prüfzeichens. <i>This test report relates to the a. m. test sample. Without permission of the test center this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any safety mark on this or similar products.</i>			

Test item particulars:	
Information about the product needed to establish a correct test program, such as product mobility, type of power connections and similar.	(Test item particulars are selected by the TRF Originator base on the requirements in the standard)
Designation	IMR18650-1500mAh, IMR18650-1200mAh
Trade mark.....	N/A
Nominal voltage.....	3.7V
Rated capacity.....	1500mAh (for IMR18650-1500mAh), 1200mAh (for IMR18650-1200mAh)
Maximum charge voltage	4.2V
End discharge voltage.....	3.0V
Manufacturer's charge method	Charging the cell with 0.2C constant current and 4.2V constant voltage until the current reduces to 0.02CmA.
Utilization Type.....	Technician replaceable cell
Sample number.....	001- 115
Possible test case verdicts:	
Test case does not apply to the test object.....	N/A
Test object does meet the requirement	P(ass)
Test object does not meet the requirement	F(ail)
Testing:	
Date of receipt of test item	2018-08-10
Date(s) of performance of tests.....	2018-08-10 to 2018-09-29
General remarks:	
This report shall not be reproduced, except in full, without the written approval of the testing laboratory.	
The test results presented in this report relate only to the object tested.	
"(see remark #)" refers to a remark appended to the report.	
"(see appended table)" refers to a table appended to the report.	
Throughout this report a point is used as the decimal separator.	

Copy of marking plate:

The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCBs that own these marks.

+ CYLINDRICAL LITHIUM-ION RECHARGEABLE CELL -

Xinxiang hongli supply source technology co., ltd

IMR18650-1500mAh DC 3.7V, 1500mAh

2018.09.01



Label of model IMR18650-1500mAh

+ CYLINDRICAL LITHIUM-ION RECHARGEABLE CELL -

Xinxiang hongli supply source technology co., ltd

IMR18650-1200mAh DC 3.7V, 1200mAh

2018.09.01



Label of Model IMR18650-1200mAh

Summary of testing:

The component cell is evaluated in this test report per the following test items according to UL 1642: 2012 Fifth Edition.

Charging method declared by the manufacturer in specifications:

ICharging the cell with 0.2CmA constant current and 4.2V constant voltage until the current reduces to 0.02CmA.

Test items:

cl.10 Short-Circuit Test;
cl.11 Abnormal Charging Test;
cl.13 CrushTest;
cl.14 ImpactTest;
cl.15 Shock Test;
cl.16 Vibration Test;
cl.17 Heating Test;
cl.18 Temperature Cycling Test;
cl.19 Low Pressure (Altitude Simulation) Test;
cl.20 Projectile Test.

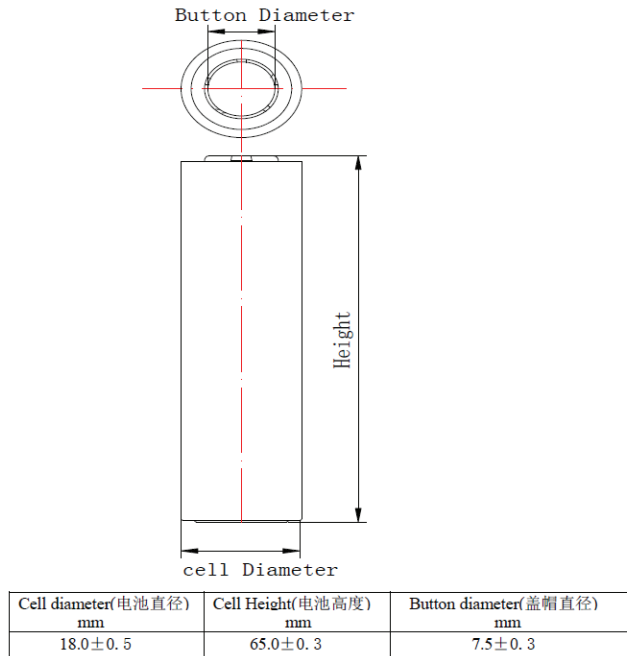
Description of the product:

This tested sample is constructed with only one single Li-ion cell and doesn't have any over current and short-circuits proof circuit.

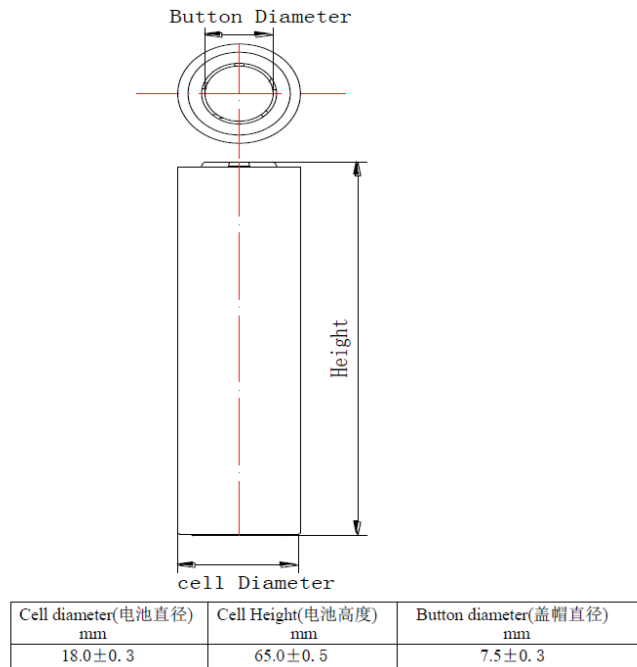
- 1) The tested cell has not been evaluated in combination with charger(s) or host product(s). Additional evaluation to determine compliance may be required on the combination(s) in the end product evaluation.
- 2) The tested cell were evaluated for a maximum charge current and maximum voltage limit outlined in the Table below. The end product evaluation shall ensure that current and voltage limits noted are maintained.
- 3) The tested cell shall be fixed in a case with sufficient protective strength as mechanical enclosure and fire enclosure. The end product shall be kept away from fire and high temperature area.

- Table 1: Electrical parameter of the cell:

Model	Nominal capacity	Nominal voltage	Nominal Charge Current	Nominal discharge current	Max. charge current	Max. discharge current	Max. charge voltage	End discharge voltage
IMR18650 -1500mAh	1500 mAh	3.7V	300mA	300mA	750mA	750mA	4.2V	3.0V
IMR18650 -1200mAh	1200 mAh	3.7V	240mA	240mA	600mA	600mA	4.2V	3.0V

Construction:

Cell 1 (IMR18650-1500mAh)



Cell 2 (IMR18650-1200mAh)

Factory location :

Xinxiang hongli supply source technology co., ltd

Dayang Village South, zhaipo Town, Xinxiang, China

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict

INTRODUCTION

1	Scope		P
2	General		P
3	Glossary		P

CONSTRUCTION

4	General		N/A
4.1	Casing		N/A
4.1.1	The casing of a lithium battery shall have the strength and rigidity necessary to resist the abuses to which it may be subjected, without resulting in a risk of fire. The casing of a user-replaceable lithium battery shall have the strength and rigidity necessary to resist the abuses to which it may be subjected without resulting in a risk of injury to persons.	Polymer lithium ion cell without enclosure	N/A
4.1.2	A cell of a user-replaceable battery shall be in a rigid casing of sufficient strength to prevent flexing. A tool providing the mechanical advantage of a pliers, screwdriver, or hacksaw shall be the minimum capable of opening the user-replaceable cell casing, if opening of the casing will expose metallic lithium.	See above	N/A
4.2	Electrolyte		N/A
4.2.1	A user-replaceable battery shall not contain pressurized vapor or liquid that could spray materials into the eyes or leak more than 5 milliliters of liquid when the battery casing is punctured under normal laboratory conditions, 23 ±2°C (73 ±3.6°F).	Technician-replaceable Batteries	N/A
4.3	Use		N/A
4.3.1	A lithium battery shall be protected from abnormal charging currents during use. A battery tested and found acceptable for the charging current, I_c (see Section 11), under fault conditions specified by the manufacturer, shall be protected from larger charging currents in the end product application by: a) Two blocking components, such as diodes, or b) One blocking component and one current limiting component, such as a resistor or a fuse. The current limiting component shall limit the charging current to one-third the value used in the Abnormal Charging Test, Section 11.	Cell only	N/A

PERFORMANCE

5	General		P
5.1	Technician-replaceable Batteries		P

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
5.1.1	Technician-replaceable lithium cells or batteries are to be tested as described in Sections 10 – 20. Section 12, Forced-Discharge Test, is applicable only to cells intended to be used in series-connected multicell applications such as battery packs. For multicell installations, also see 5.3.1.		P
5.1.2	When a fire or explosion occurs as a result of the Crush Test, Section 13, or the Impact Test, Section 14, or the cell or battery ruptures to the extent that the aluminum test cage is penetrated during the Projectile Test, Section 20; the use of the technician replaceable cell or battery shall be restricted to applications in which it is not exposed to, or is protected from, any conditions shown to cause a fire or explosion.	No fire or explosion occurs	P
5.1.3	Cells and batteries subjected to the Shock Test, Section 15, Vibration Test, Section 16, Temperature Cycling Test, Section 18, and Low Pressure (Altitude Simulation) Test, Section 19, shall also not leak or vent. For these tests, unacceptable leakage is determined to have occurred when the resulting mass loss exceeds the values shown in Table 5.1, Venting and leakage mass loss criteria.	No leak or vent	P
5.2	User-replaceable Batteries		N/A
5.2.1	User-replaceable lithium cells or batteries are to be tested as described in Sections 10-20. Section 12, Forced Discharge Test, is applicable only to cells intended to be used in multicell applications such as battery packs. In addition to complying with the requirements for a technician replaceable cell or battery as specified in 5.1.1, a user-replaceable cell or battery shall not explode or ignite when subjected to the Crush Test, Section 13, or the Impact Test, Section 14.	Technician-replaceable Cell	N/A
5.2.2	Sets of five specimens each are to be used for the Projectile Test, Section 20.3; see Table 6.3. When only one specimen from a set of five does not comply with the requirements, another set of five specimens is to be tested. All specimens from this second set shall comply with the requirements.	Technician-replaceable Cell	N/A
5.3	Multi-cell Installation		N/A
5.3.1	A technician-replaceable or user-replaceable cell intended for use in multicell installations or battery packs shall also be tested as described in 10.3 and Section 12. No fire or explosion shall occur as a result of these tests. In addition, batteries subjected to the test described in 10.3 shall meet the requirements as described in 5.1.1 and 5.2.1 for a cell or battery subjected to the Short-Circuit Test, Section 10.	Single cell	N/A

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
6	Samples		P
6.1	Fully charged primary cells or batteries and primary cells or batteries that have been conditioned by partial or complete discharge, or both, are to be used for the tests described in Sections 10-20. The number of samples to be used in each test for a primary technician-replaceable cell or battery is shown in Table 6.1. The number of samples to be used in each test for a primary user-replaceable cell or battery is shown in Table 6.3. When a group of cells or batteries of different sizes, but similar chemistries is involved, selected sizes representative of the range are to be tested.	The samples are secondary lithium-ion cells.	N/A
6.2	Fully charged secondary cells or batteries and secondary cells or batteries that have been conditioned by charge-discharge cycling are to be used for the tests described in Sections 10 – 20. The number of samples to be used in each test for a secondary technician-replaceable cell or battery is shown in Table 6.2. The number of samples to be used in each test for a secondary user-replaceable cell or battery is shown in Table 6.4. When a group of cells or batteries of different sizes, and similar chemistries is involved, selected sizes representative of the range are to be tested.	Prepared as required.	P
6.3	Prior to conducting the testing in Section 17, the lithium ion cell samples shall be pre-conditioned as outlined in 6.4 and 6.5.	Prepared as required.	P
6.4	For the heating test of Section 17, two sets of five lithium ion cell samples are to be fully discharged (i.e. to the manufacturer's specified end point voltage). The samples are then placed in a test chamber and conditioned for 1 to 4 h (5 samples at the upper temperature limit and 5 samples at the lower temperature limit of the operating region) as outlined in Table 6.3.	Prepared as required.	P
6.5	While still in the test chamber set at the temperature limits, the samples are charged (5 samples at the upper temperature limit and 5 samples at lower temperature limit) at the specified maximum charging current and upper limit charging voltage per Table 6.3, using a constant voltage charging method. Charging is continued until the charge current is reduced to the specified end of charge conditions (i.e. 0.05 times the charge current).	Prepared as required.	P
7	Conditioning of Samples		P
7.1	Discharge		N/A

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
7.1.1	Primary batteries are to be completely discharged by connecting their terminals through resistors that provide the desired level of discharge within 60 days. Completely discharged is considered to be the state in which the closed-circuit voltage has been reduced to less than 0.2 volts and the short-circuit current to less than 1.0 milliamperes. Batteries are to be discharged at room temperature. Cells with a liquid cathode such as thionyl chloride or sulfur dioxide, shall also be conditioned by one-half discharge	The samples are secondary lithium-ion cells.	N/A
7.1.2	For solid electrolyte and other types of primary lithium batteries that cannot be discharged within 60 days because of the small currents they inherently produce, longer discharge times plus discharge at higher temperatures may be used to obtain the desired level of discharge. The manufacturer's recommended discharge procedures are to be followed so as to obtain the required discharge level in the minimum time.	The samples are secondary lithium-ion cells.	N/A
7.2	Charge-discharge cycling		P
7.2.1	Secondary cells are to be conditioned at 25°C (77°F). Cells are continuously cycled as per the manufacturer's specifications. The specification shall be such that the full rated capacity of the cell is utilized and the number of cycles accumulated shall be at least equal to 25 percent of the advertised cycle life of the cell or cycled continuously for 90 days, whichever is shorter. Cycling is to be done either individually or in groups. Cells are to be recharged prior to testing as indicated in Table 6.2 and Table 6.4.	The samples are cycled at manufacturer's factory before they were sent for test.	P
8	Important test considerations		P
8.1	Some lithium batteries are capable of exploding when the tests described in Sections 10-20 are conducted. It is important that personnel be protected from the flying fragments, explosive force, sudden release of heat, and noise that results from such explosions. The test area is to be well ventilated to protect personnel from possible harmful fumes or gases.	Prepared the tests as required.	P
8.2	As an additional precaution, the temperatures on the surface of the battery casings shall be monitored during the tests described in Sections 10, 12, 13, and 14. All personnel involved in the testing of lithium batteries are to be instructed never to approach a lithium battery while the surface temperature exceeds 90°C (194°F).	Prepared the tests as required.	P
8.3	For protection, the Projectile Test, Section 20 is to be conducted in a room separate from the observer.	Prepared the tests as required.	P

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
9	Temperature measurements		P
9.1	Temperatures are to be measured by thermocouples consisting of wires not larger than 24 AWG (0.21 mm ²) and not smaller than 30 AWG (0.05 mm ²) and a potentiometer-type instrument.	Prepared the tests as required.	P
9.2	The temperature measurements on the batteries are to be made with the measuring junction of the thermocouple held tightly against the metal casing of the battery.	Prepared the tests as required.	P
TESTS FOR TECHNICIAN-REPLACEABLE AND USER-REPLACEABLE BATTERIES			
ELECTRICAL TESTS			
10	Short-Circuit Test		P
10.1	Each test sample battery, in turn, is to be short-circuited by connecting the positive and negative terminals of the battery with a circuit load having a maximum resistance load of 0.1 ohm. The battery is to discharge until a fire or explosion is obtained, or until it has reached a completely discharged state of less than 0.1 volts and the battery case temperature has returned to $\pm 10^{\circ}\text{C}$ ($\pm 18^{\circ}\text{F}$) of ambient temperature.	Tested as required.	P
10.2	Tests are to be conducted at $20 \pm 5^{\circ}\text{C}$ ($68 \pm 9^{\circ}\text{F}$) and at $55 \pm 5^{\circ}\text{C}$ ($131 \pm 9^{\circ}\text{F}$). The batteries are to reach equilibrium at $20 \pm 5^{\circ}\text{C}$ or $55 \pm 5^{\circ}\text{C}$, as applicable, before the terminals are connected.	Tested as required.	P
10.3	A battery is to be tested individually unless the manufacturer indicates that it is intended for use in series or parallel. For series or parallel use, additional tests on five sets of batteries are to be conducted using the maximum number of batteries to be covered for each configuration.	Tested as required.	P
10.4	When an overcurrent or thermal protective device that has been investigated for the purpose actuates during the test, the test shall be repeated with the battery supply connected to the maximum load that does not cause the protective device to open. A protective device that has not been investigated for the purpose shall be short-circuited.	Only one single lithium-ion cell, no over-current or thermal protective device was integrated into the cell.	N/A
10.5	The samples shall not explode or catch fire. The temperature of the exterior cell or battery casing shall not exceed 150°C (302°F).	The test results meet the requirements.	P
11	Abnormal Charging Test		P
11.1	Primary cells or batteries shall comply with 11.2—11.7.	Secondary cell	N/A
11.2	Cells or batteries conditioned in accordance with Tables 6.1 or 6.3, as applicable, are to be used for this test. The batteries are to be tested in an ambient temperature of $20 \pm 5^{\circ}\text{C}$ ($68 \pm 9^{\circ}\text{F}$).		N/A

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
11.3	Each test sample battery is to be subjected to a charging current of three times the current I, specified by the manufacturer by connecting it in opposition to a dc-power supply. The specified charging current is to be obtained by connecting a resistor of the specified size and rating in series with the battery. The test charging time is to be calculated using the formula: $t_c = \frac{2.5C}{3(I_c)}, \text{ in which}$ t_c is the charging time in hour C is the capacity of the cell/battery in ampere-hours, and I_c is the maximum charging current, in amperes, specified by the manufacturer. The minimum charging time is to be 7 hours.		N/A
11.4	When a non-resettable overcurrent or thermal protective device that has been investigated for the purpose operates during the test, the test is to be repeated at a charge current below the level that the protective device operates. When a resettable protective device operates during the test, the protector is allowed to reset to a total of 10 cycles; or until the appropriate charging time has been completed, but not less than 7 hours. A protective device that has not been investigated for the purpose is to be short-circuited. See 2.3.2.		N/A
11.5	The samples shall not explode or catch fire.		N/A
11.6	Secondary cells or batteries shall comply with 11.7— 11.10.	See table 11, tested as required	P
11.7	Cells or batteries conditioned in accordance with Tables 6.2 or 6.4, as applicable, are to be used for this test. The batteries are to be tested in an ambient temperature of $20 \pm 5^\circ\text{C}$ ($68 \pm 9^\circ\text{F}$).	Tested as required.	P
11.8	Each test sample battery is to be discharged at a constant current of $0.2C/\text{hour}$, to a manufacturer specified discharge endpoint voltage. The cell or battery is then to be charged with a constant maximum specified output voltage and a current limit of three times the maximum charging current, I , specified by the manufacturer. Charging duration is to be 7 hours or the time required to reach the manufacturer's specified end-of-charge condition, whichever is greater.	Tested as required.	P

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
11.9	The cell/battery is to be tested without the assistance of overcurrent or thermal protective devices, unless such protective devices have been investigated for the purpose. When a non-resettable overcurrent or thermal protective device operates during the test, the test shall be repeated at an overcharging current below the level that the protection device operates. When a resettable protective device operates during the test, the protector is to be allowed to reset to a total of 10 cycles; or until the appropriate charging time has been completed, but not less than 7 hours. A protective device that has not been investigated for the purpose is to be short-circuited. See 2.3.2.	The samples are to be tested without any assistance of over-current or thermal protective devices.	P
11.10	The samples shall not explode or catch fire.	No explosion or catch fire during and after the test	P
12	Forced-Discharged Test		N/A
12.1	This test is intended for cells that are to be used in series-connected, multicell applications, such as battery packs.	One single cell	N/A
12.2	A completely discharged cell is to be force-discharged by connecting it in series with fully charged cells of the same kind. The number of fully charged cells to be connected in series with the discharged cell is to equal the maximum number less one of the cells to be covered for series use. Five cells are to be completely discharged, at room temperature.		N/A
12.3	Once the completely discharged cell is connected in series with the specified number of fully charged cells the resultant battery pack is to be short circuited.		N/A
12.4	The positive and negative terminals of the sample are to be connected with a copper wire with a maximum resistance load of 0.1 ohm. The sample is to discharge until a fire or explosion is obtained, or until it has reached a completely discharged state of less than 0.2 volts and the battery case temperature has returned to $\pm 10^{\circ}\text{C}$ (18°F) of ambient temperature.		N/A
12.5	When an overcurrent or thermal protective device that has been investigated for the purpose operates during the test, the test shall be repeated with the battery supply connected to the maximum load that does not cause the protective device to open. A protective device that has not been investigated for the purpose shall be short-circuited.		N/A
12.6	The samples shall not explode or catch fire.		N/A
MECHANICAL TESTS			

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
13	Crush Test		P
13.1	A battery is to be crushed between two flat surfaces. The force for the crushing is to be applied by a hydraulic ram or similar force mechanism. The flat surfaces are to be brought in contact with the cells and the crushing is to be continued until an applied force of 13 ± 1 kN (3000 ± 224 lbs) is reached. Once the maximum force has been obtained it is to be released.	Tested as required.	P
13.2	A cylindrical, pouch or prismatic cell is to be crushed with its longitudinal axis parallel to the flat surfaces of the crushing apparatus. A prismatic cell is also to be rotated 90° around its longitudinal axis so that both the wide and narrow sides will be subjected to the crushing force. Each sample is to be subjected to a crushing force in only one direction. Separate samples are to be used for each test. Exception: For Lithium ion systems, a cylindrical, pouch or prismatic cell is to be crushed with its longitudinal axis parallel to the flat surface of the crushing apparatus. Each sample is to be subjected to a crushing force in only one direction. Test only the wide side of pouch and prismatic cells.	Tested as required.	P
13.3	A coin or button battery is to be crushed with the flat surface of the battery parallel with the flat surfaces of the crushing apparatus.	Not a coin or button battery	N/A
13.4	The samples shall not explode or catch fire.	No explosion or catch fire during and after the test	P
14	Impact Test		P
14.1	A test sample battery is to be placed on a flat surface. A 15.8 ± 0.1 -mm ($5/8 \pm 0.004$ -in) diameter bar is to be placed across the center of the sample. A 9.1 ± 0.46 -kg (20 ± 1 -lb) weight is to be dropped from a height of 610 ± 25 mm (24 ± 1 in) onto the sample.	Prepared the test as required.	P

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
14.2	A cylindrical, pouch or prismatic cell is to be impacted with its longitudinal axis parallel to the flat surface and perpendicular to the longitudinal axis of the 15.8-mm (5/8-in) diameter curved surface lying across the center of the test sample. A prismatic cell is also to be rotated 90° around its longitudinal axis so that both the wide and narrow sides are subjected to the impact. Each sample is to be subjected to only a single impact. Separate samples are to be used for each test. Exception: For Lithium ion systems, a cylindrical, pouch or prismatic cell is to be impacted with its longitudinal axis parallel to the flat surface and perpendicular to the longitudinal axis of the 15.8-mm (5/8-in) diameter curved surface lying across the center of the test sample. Each sample is to be subjected to only a single impact. Test only the wide side of pouch and prismatic cells.	Tested as required.	P
14.3	A coin or button battery is to be impacted with the flat surface of the test sample parallel to the flat surface and the 15.8-mm (5/8-in) diameter curved surface lying across its center.	Not a coin or button battery	N/A
14.4	The samples shall not explode or catch fire.	No explosion or catch fire during and after the test	P
15	Shock Test		P
15.1	The cell is to be secured to the testing machine by means of a rigid mount which supports all mounting surfaces of the cell. Each cell shall be subjected to a total of three shocks of equal magnitude. The shocks are to be applied in each of three mutually perpendicular directions unless it has only two axes of symmetry in which case only two directions shall be tested. Each shock is to be applied in a direction normal to the face of the cell. For each shock the cell is to be accelerated in such a manner that during the initial 3 milliseconds the minimum average acceleration is 75 g (where g is the local acceleration due to gravity). The peak acceleration shall be between 125 and 175 g. Cells shall be tested at a temperature of 20 ± 5°C (68 ± 9°F).	Prepared the test as required. See table 15	P
15.2	The samples shall not explode or catch fire. In addition, the sample shall not vent or leak as described in 5.1.1.	No explosion or catch fire, the sample not vent or leak	P
16	Vibration Test		P
16.1	A battery is to be subjected to simple harmonic motion with an amplitude of 0.8 mm (0.03 inch) [1.6 mm (0.06 inch) total maximum excursion].	Prepared the test as required. See table 16.	P

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
16.2	The frequency is to be varied at the rate of 1 hertz per minute between 10 and 55 hertz, and return in not less than 90 nor more than 100 minutes. The battery is to be tested in three mutually perpendicular directions. For a battery that has only two axes of symmetry, the battery is to be tested perpendicular to each axis.	Tested as required.	P
16.3	The samples shall not explode or catch fire. In addition the sample shall not vent or leak as described in 5.1.1.	The test results meet the requirements.	P
ENVIRONMENTAL TESTS			
17	Heating Test		P
17.1	A battery is to be heated in a gravity convection or circulating air oven with an initial temperature of $20 \pm 5^{\circ}\text{C}$ ($68 \pm 9^{\circ}\text{F}$). The temperature of the oven is to be raised at a rate of $5 \pm 2^{\circ}\text{C}$ ($9 \pm 3.6^{\circ}\text{F}$) per minute to a temperature of $130 \pm 2^{\circ}\text{C}$ ($266 \pm 3.6^{\circ}\text{F}$) and remain for 10 min. The sample shall return to room temperature ($20 \pm 5^{\circ}\text{C}$) and then be examined. For batteries specified for temperatures above 100°C (212°F), the conditioning temperature shall be increased from $130 \pm 2^{\circ}\text{C}$ ($266 \pm 3.6^{\circ}\text{F}$), to $30 \pm 2^{\circ}\text{C}$ ($86 \pm 3.6^{\circ}\text{F}$) above the manufacturers maximum specified temperature. For a battery of lithium metal chemistry, the conditioning temperature shall be increased to a maximum of $170 \pm 2^{\circ}\text{C}$ ($338 \pm 3.6^{\circ}\text{F}$).	Tested as required.	P
17.2	The samples shall not explode or catch fire.	The test results meet the requirements.	P
18	Temperature Cycling Test		P
18.1	The batteries are to be placed in a test chamber and subjected to the following cycles: Raising the chamber-temperature to $70 \pm 3^{\circ}\text{C}$ ($158 \pm 5^{\circ}\text{F}$) within 30 minutes and maintaining this temperature for 4 hours. Reducing the chamber temperature to $20 \pm 3^{\circ}\text{C}$ ($68 \pm 5^{\circ}\text{F}$) within 30 minutes and maintaining this temperature for 2 hours. Reducing the chamber temperature to minus $40 \pm 3^{\circ}\text{C}$ (minus $40 \pm 5^{\circ}\text{F}$) within 30 minutes and maintaining this temperature for 4 hours. Raising the chamber temperature to $20 \pm 3^{\circ}\text{C}$ ($68 \pm 5^{\circ}\text{F}$) within 30 minutes. Repeating the sequence for a further 9 cycles. After the 10th cycle, storing the batteries for a minimum of 24 hours, at a temperature of $20 \pm 5^{\circ}\text{C}$ ($68 \pm 9^{\circ}\text{F}$) prior to examination.	Tested as required. See table 18	P
18.2	The samples shall not explode or catch fire. In addition, the samples shall not vent or leak as described in 5.1.1.	The test results meet the requirements.	P
19	Low Pressure (Altitude Simulation) Test		P

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
19.1	Sample batteries are to be stored for 6 hours at an absolute pressure of 11.6 kPa (1.68 psi) and a temperature of $20 \pm 3^{\circ}\text{C}$ ($68 \pm 5^{\circ}\text{F}$).	Tested as required. See table 19	P
19.2	The samples shall not explode or catch fire as a result of the Altitude Simulation Test. In addition, the samples shall not vent or leak as described in 5.1.1.	The test results meet the requirements.	P
TESTS FOR USER-REPLACEABLE LITHIUM BATTERIES			
20	Projectile Test		P
20.1	When subjected to the test described in 20.2 - 20.5 no part of an exploding cell or battery shall penetrate the wire screen such that some or all of the cell or battery protrudes through the screen.	The test results meet the requirements.	P
20.2	Each test sample cell or battery is to be placed on a screen that covers a 102 mm (4 inch) diameter hole in the center of a platform table. The screen is to be constructed of steel wire mesh having 20 openings per inch (25.4 mm) and a wire diameter of 0.017 inch (0.43 mm).	Prepared the test as required.	P
20.3	The screen is to be mounted 38 mm (1-1/2 inch) above a burner. The fuel and air flow rates are to be set to provide a bright blue flame that causes the supporting screen to glow a bright red.	Prepared the test as required.	P
20.4	An eight-sided covered wire cage, 610 mm (2 feet) across and 305 mm (1 foot) high, made from metal screening is to be placed over the test sample. See Figure 20.1. The metal screening is to be constructed from 0.25 mm (0.010 inch) diameter aluminum wire with 16 — 18 wires per inch (25.4 mm) in each direction.	Tested as required.	P
20.5	The sample is to be heated and shall remain on the screen until it explodes or the cell or battery has ignited and burned out. It is not required to secure the sample in place unless it is at risk of falling off the screen before the test is completed. When required, the sample shall be secured to the screen with a single wire tied around the sample.	Tested as required.	P
MARKING			
21	General		P
21.1	A battery shall be legibly and permanently marked with: The manufacturer's name, trade name, or trademark or other descriptive marking by which the organization responsible for the product may be identified; A distinctive ("catalog" or "model") number or the equivalent; The date or other dating period of manufacture not exceeding any three consecutive months.	See marking plate on page 3	P

UL1642: 2012			
Clause	Requirement + Test	Result - Remark	Verdict
21.2	If a manufacturer produces a battery at more than one factory, each battery package shall have a distinctive marking to identify it as the product of a particular factory.	One single factory	N/A

Critical Components

Material: e.g. external enclosure, PCB, closed-end connector, sleeves, cord anchorage etc

Components with winding: e.g. motor, transformer, magnetic coil etc.

Other components: e.g. switch, thermostat, heater, plug, internal wire, capacitor, relay, varistor etc.

Object/ Part No.	Manufacturer/ trademark	Type/ Model	Technical data	Standard	Mark(s) of conformity
Cell	Henan Huiqiang Science and Technology New Energy materials Technology Co. LTD	IMR18650- 1500mAh, IMR18650- 1200mAh	DC 3.7V, 1500mAh (for IMR18650-1500mAh), DC 3.7V, 1200mAh (for IMR18650-1200mAh)	UL 1642	Tested with appliance
-Electrolyte	Henan Farnwright New Energy Technology Co. LTD	HL-01	LiPF ₆ +DEC+EC+EMC, Conductivity: 10.5mS/cm, H ₂ O<20ppm, HF<50ppm	--	--
-Separator	Henan Huiqiang Science and Technology New Energy materials Technology Co. LTD	PP	1200mm×0.025mm×60mm, Shutdown temperature: 130°C	--	--
-Positive electrode	Xinxiang Hongli Power Supply Technology co. LTD	01	LiMn ₂ O ₄ , Carbon black, PVDF, Conductive additive, Al	--	--
-Negative electrode	Xinxiang Hongli Power supply Technology Co. LTD	A138	Graphite, CMC, SBR, Distilled Water	--	--

10	TABLE: Short-Circuit Test					P
Fully Charged Cell						
Ambient temperature: 23.2°C						
Sample No.	001*	002*	003*	004*	005*	
Tmax(°C)	79.8	81.6	84.0	77.5	79.3	
Failure Mode	No	No	No	No	No	
Ambient temperature: 23.7°C						
Sample No.	006*	007*	008*	009*	010*	
Tmax(°C)	77.3	81.4	78.9	77.4	83.9	
Failure Mode	No	No	No	No	No	
Cycled Cell						
Ambient temperature: 57.9°C						
Sample No.	011*	012*	013*	014*	015*	
Tmax(°C)	87.7	88.9	92.1	86.8	85.4	
Failure Mode	No	No	No	No	No	
Ambient temperature: 56.9°C						
Sample No.	016*	017*	018*	019*	020*	
Tmax(°C)	87.8	83.1	91.7	83.7	84.9	
Failure Mode	No	No	No	No	No	
Fully Charged Cell						
Ambient temperature: 24.0°C						
Sample No.	001**	002**	003**	004**	005**	
Tmax(°C)	72.3	79.5	78.7	77.2	79.2	
Failure Mode	No	No	No	No	No	
Ambient temperature: 23.6°C						
Sample No.	006**	007**	008**	009**	010**	
Tmax(°C)	79.4	77.3	79.0	77.4	78.0	
Failure Mode	No	No	No	No	No	
Cycled Cell						
Ambient temperature: 57.9°C						
Sample No.	011**	012**	013**	014**	015**	
Tmax(°C)	89.2	90.4	89.4	89.5	90.9	
Failure Mode	No	No	No	No	No	
Ambient temperature: 56.7°C						
Sample No.	016**	017**	018**	019**	020**	

Tmax(°C)	89.4	86.1	82.1	87.6	86.6
Failure Mode	No	No	No	No	No
Supplementary information: Tmax was recorded on the centre of the outside case. No explode or catch fire. The temperature of the cell casing does not exceed 150°C. Remark: *Tested with IMR18650-1200mAh. **Tested with IMR18650-1500mAh.					

11	TABLE: Abnormal Charging Test				P
Ambient temperature: 24.1°C					
Id	0.24 A		Ue	3.0 V	
Ic	0.6 A		Uc	4.2 V	
Fully Charged Cell					
Sample No.	021*	022*	023*	024*	025*
Tmax(°C)	25.1	24.6	24.7	24.9	25.3
Failure Mode	No	No	No	No	No
Cycled cell*					
Sample No.	026*	027*	028*	029*	030*
Tmax(°C)	25.3	24.9	24.8	24.6	25.1
Failure Mode	No	No	No	No	No
Ambient temperature: 24.1°C					
Id	0.3 A		Ue	3.0 V	
Ic	0.75 A		Uc	4.2 V	
Fully Charged Cell					
Sample No.	021**	022**	023**	024**	025**
Tmax(°C)	24.4	24.7	24.9	25.6	24.8
Failure Mode	No	No	No	No	No
Fully Charged Cell					
Sample No.	026**	027**	028**	029**	030**
Tmax(°C)	25.5	24.8	24.6	24.9	25.1
Failure Mode	No	No	No	No	No
Supplementary information: Tmax was recorded on the centre of the outside case; No explode or catch fire. Remark: *Tested with IMR18650-1200mAh. **Tested with IMR18650-1500mAh.					

**Tested with IMR18650-1500mAh.

14	TABLE: Impact Test					P
Ambient temperature: 23.8°C						
Fully Charged Cell						
Sample No.	041*	042*	043*	044*	045*	
Tmax(°C)	112.5	108.5	114.3	109.2	105.4	
Impact direction	Wide side	Wide side	Wide side	Wide side	Wide side	
Failure mode	No	No	No	No	No	
Cycled cell						
Sample No.	046*	047*	048*	049*	050*	
Tmax(°C)	110.7	107.5	108.6	106.9	110.0	
Impact direction	Wide side	Wide side	Wide side	Wide side	Wide side	
Failure mode	No	No	No	No	No	
Fully Charged Cell						
Sample No.	041**	042**	043**	044**	045**	
Tmax(°C)	109.8	106.6	109.5	102.8	113.7	
Impact direction	Wide side	Wide side	Wide side	Wide side	Wide side	
Failure mode	No	No	No	No	No	
Cycled cell						
Sample No.	046**	047**	048**	049**	050**	
Tmax(°C)	109.3	112.8	106.3	110.9	109.0	
Impact direction	Wide side	Wide side	Wide side	Wide side	Wide side	
Failure mode	No	No	No	No	No	
Supplementary information: Cylindrical cell, no explosion or catch fire. Remark: *Tested with IMR18650-1200mAh. **Tested with IMR18650-1500mAh.						

15	TABLE: Shock Test					P
Ambient temperature: 22.7°C						
Fully Charged Cell						
Sample No.	051*	052*	053*	054*	055*	
Mass before test (g)	35.859	36.012	36.078	35.984	35.886	
Mass after test (g)	35.857	36.010	36.077	35.983	35.886	
Cycled cell						
Sample No.	056*	057*	058*	059*	060*	
Mass before test (g)	36.102	36.045	35.784	35.882	36.106	
Mass after test (g)	36.102	36.044	35.782	35.880	36.105	

Fully Charged Cell					
Sample No.	051**	052**	053**	054**	055**
Mass before test (g)	39.227	39.562	39.855	39.426	39.805
Mass after test (g)	39.225	39.561	39.852	39.425	39.802
Cycled cell**					
Sample No.	056**	057**	058**	059**	060**
Mass before test (g)	39.524	39.906	38.897	39.212	38.995
Mass after test (g)	39.522	39.905	38.894	39.210	38.993
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%					
Remark:					
*Tested with IMR18650-1200mAh.					
**Tested with IMR18650-1500mAh.					

16	TABLE: Vibration Test				P
Ambient temperature: 23.2°C					
Fully Charged Cell					
Sample No.	061*	062*	063*	064*	065*
Mass before test (g)	35.992	36.036	35.997	35.896	36.107
Mass after test (g)	35.991	36.035	35.995	35.893	36.106
Cycled cell*					
Sample No.	066*	067*	068*	069*	070*
Mass before test (g)	36.112	36.025	36.114	35.988	35.845
Mass after test (g)	36.111	36.023	36.111	35.987	35.843
Fully Charged Cell					
Sample No.	061**	062**	063**	064**	065**
Mass before test (g)	39.233	38.892	39.056	39.115	39.217
Mass after test (g)	39.231	38.890	39.055	39.113	39.215
Cycled cell**					
Sample No.	066**	067**	068**	069**	070**
Mass before test (g)	39.336	39.102	38.976	39.347	39.118
Mass after test (g)	39.334	39.101	38.973	39.344	39.117
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1%					
Remark:					
*Tested with IMR18650-1200mAh.					
**Tested with IMR18650-1500mAh.					

19	TABLE: Low Pressure (Altitude Simulation) Test				P
Ambient temperature: 22.9°C					
Fully Charged Cell					
Sample No.	101*	102*	103*	104*	105*
Mass before test (g)	35.982	36.103	36.132	36.069	35.921
Mass after test (g)	35.980	36.101	36.130	36.068	35.919
Cycled cell					
Sample No.	106*	107*	108*	109*	110*
Mass before test (g)	35.715	36.172	36.155	35.954	36.221
Mass after test (g)	35.713	36.171	36.153	35.952	36.220
Fully Charged Cell					
Sample No.	101**	102**	103**	104**	105**
Mass before test (g)	39.221	39.528	39.267	39.455	38.893
Mass after test (g)	39.219	39.526	39.266	39.453	38.892
Cycled cell					

Sample No.	106**	107**	108**	109**	110**
Mass before test (g)	39.113	39.906	39.548	38.698	38.806
Mass after test (g)	39.111	39.904	39.546	38.695	38.786
Supplementary information: no explosion or catch fire, in addition the sample did not vent or leak. Max loss less than 0.1% Remark: *Tested with IMR18650-1200mAh. **Tested with IMR18650-1500mAh.					

--End of report--

Produkte
Products

	Eq. No.	Name	Manufacturer	Model No.	Date of Calibration	Date of next Calibration
☒	E009	Thermal cycling chamber	DONGGUAN HUAYIN INSTRUCTION CO., LTD	WGD/JS-80L	2018.10.22	2019.10.21
☒	E010	Electric constant temperature drying oven	Shanghai Keelrein Scientific Instruction Co., Ltd.	DHG-9240A	2018.10.22	2019.10.21
☒	E011	Electric constant temperature drying oven	Shanghai Keelrein Scientific Instruction Co., Ltd.	DHG-9240A	2018.10.22	2019.10.21
☒	E018	Vernier caliper	Mahr GmbH Esslingen	MarCal 16EX	2018.10.22	2019.10.21
☒	E020	Projectile test system	DONGGUAN HUAYIN INSTRUCTION CO., LTD	HY-RS-02	2017.11.29	2018.11.29
☒	E021	Temperature and Humidity Data Logger	Shanghai Meteorological Instrument Factory Co., Ltd.	ZJ1-2B	2017.12.03	2018.12.02
☒	E022	Stopwatches	ShenZhen HUIBO Industrial & Trading Co., Ltd	PC894	2018.10.23	2019.10.21
☒	E045	Thermal cycling chamber	MCM TECHNOLOGY (HONG KONG) LIMITED	MCM-80L	2017.11.30	2018.11.29
☒	E050	Performance testing device of secondary battery	ShenzhenNeware Technology Limited	CT-3004W-20V10A-NTF	2018.02.10	2019.02.09
☒	E061	Low pressure chamber	DONGGUAN HUAYIN INSTRUCTION CO., LTD	HY-ZK-1A	2018.03.24	2019.03.23
☒	E069	Electronic load	ITECH ELECTRONICS	HT3300	2018.03.26	2019.03.25
☒	E086	Impact Tester	MCM TECHNOLOGY (HONG KONG) LIMITED	0~1.5m	2018.06.13	2019.06.12
☒	E102	DC power	LONGWEI INSTRUMENTS (HK) CO., LTD	LW-3020KD	2018.08.27	2019.08.26
☒	E107	Performance testing device of secondary battery	Shenzhen Neware Technology Limited Company	CT-3008-10V6A-A	2017.12.29	2018.12.28
☒	E108	Performance testing device of secondary battery	Shenzhen Neware Technology Limited Company	CT-3008-10V6A-A	2018.05.11	2019.05.10
☒	E110	Electro-dynamic vibration test system	SUZHOUS FNS VIBRATION SYSTEM CO., LTD	DC-1000-12/TB0505	2018.01.05	2019.01.04
☒	E112	Performance testing device of secondary battery	ShenzhenNeware Technology Limited	CT-4008-10V6A-FA	2018.05.19	2019.05.18
☐	E113	Performance testing device of secondary battery	ShenzhenNeware Technology Limited	CT-4008-5V0.1A-164	2018.05.19	2019.05.18
☒	E120	Multi channel temperature recorder	AGILENT TECHNOLOGIES	34970A	2018.03.28	2019.03.27
☒	E125	Multimeter	Fluke Incorporation	FLUKE17B	2018.08.24	2019.08.23
☒	E135	Hydraulic shock systems	SUZHOUS FNS VIBRATION SYSTEM CO., LTD	SAIS-10	2017.11.22	2018.11.21
☒	E138	Performance testing device of secondary battery	ShenzhenNeware Technology Limited	CT-3008W-5V6A-S1	2017.11.17	2018.11.16
☒	E148	Electronic balance	SHANHAI LIANG PING INSTRUCTION CO., LTD	JA5003	2017.11.29	2018.11.28
☒	E150	Steel tape	MCM TECHNOLOGY (HONG KONG) LIMITED	(0-5)m	2017.11.23	2018.11.22
☒	E167	Ground Resistance Tester	Zhuhai Anlian Electronics Co.,Ltd.	AN9616H	2018.01.16	2019.01.15
☒	E234	Crusher test system	Shenzhen Demaisheng Measuring&Controlling Equipment Co.,Ltd.	DMS-9981	2018.08.29	2019.08.28

*) Initial calibration or verification only

Statement of Uncertainty

Unless otherwise specified, combined measurement uncertainty for values stated in the test report is as stated below:

Voltage measurement: $\pm 1.50\%$ (true rms value)

Current measurement:	±0.05% (DC voltage)
	±1.56% (true rms value)
	±0.30% (DC current)
Frequency	±0.06%
Resistance - between 20 mΩ and 1 MΩ	±0.21%
- another values	±2.00%
Temperature- below 100°C	±0.5°C (without thermocouple; for thermocouple add 2°C)
- between 100 and 500°C	±0.6°C (without thermocouple; for thermocouple add 2°C)
Time - below 20 s	±0.5%
- more than 20 s (manual meas.)	±0.2 s
Linear dimensions - less than 1 mm	±0.01 mm
- from 1 to 25 mm	±0.05 mm
- more than 25 mm	±0.60%
Mass - below 5 kg	±1%
- 5 kg and more	±0.05%
Force	±1%
Relative humidity	±5%
Air pressure (barometric)	±0.8kPa
Pressure	±3.0%

Values stated in this document represent the worst case for equipment which is in possession of the laboratory and setups commonly used for testing.

For units or cases not specified in this document the evaluation of uncertainty shall be made upon request on individual basis.

The reported combined uncertainty is stated as standard uncertainty of reported value multiplied by coverage factor $k = 2$, which for normal distribution corresponds to a coverage probability of approximately 95%.

Product: CYLINDRICAL LITHIUM-ION RECHARGEABLE CELL

Type Designation: IMR18650-1500mAh, IMR18650-1200mAh

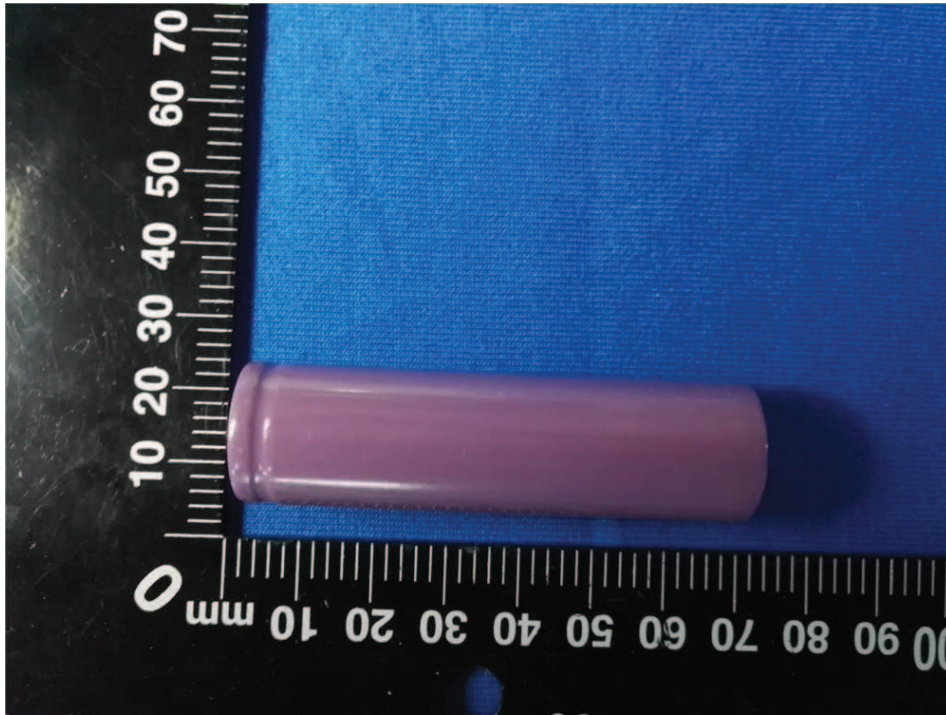


Figure 1 Front view of cell (model: IMR18650-1200mAh)

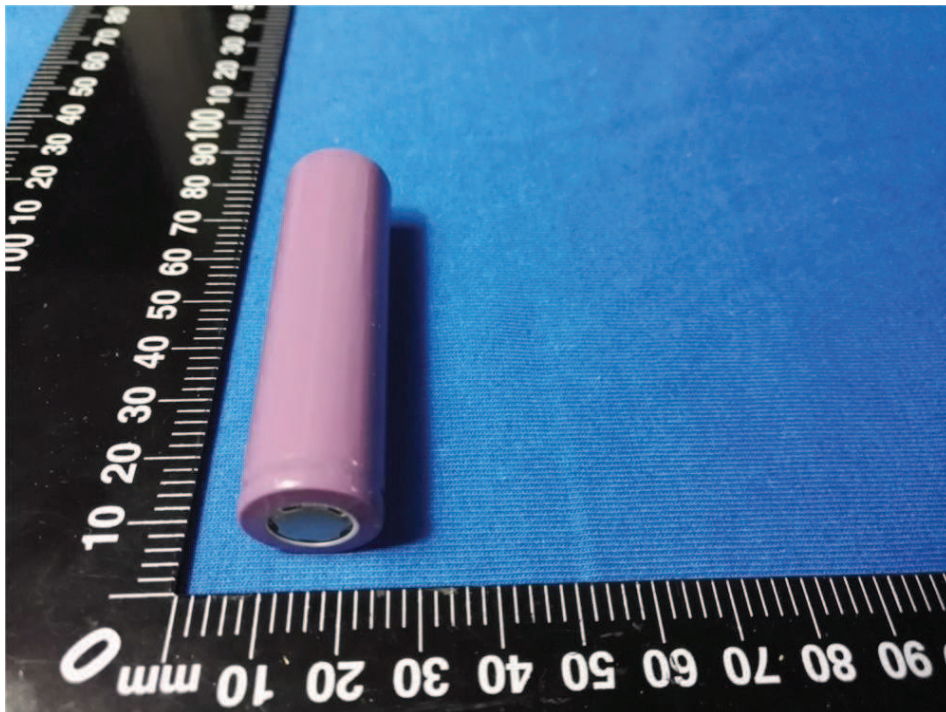


Figure 2 Back view of cell (model: IMR18650-1200mAh)

Product: CYLINDRICAL LITHIUM-ION RECHARGEABLE CELL

Type Designation: IMR18650-1500mAh, IMR18650-1200mAh

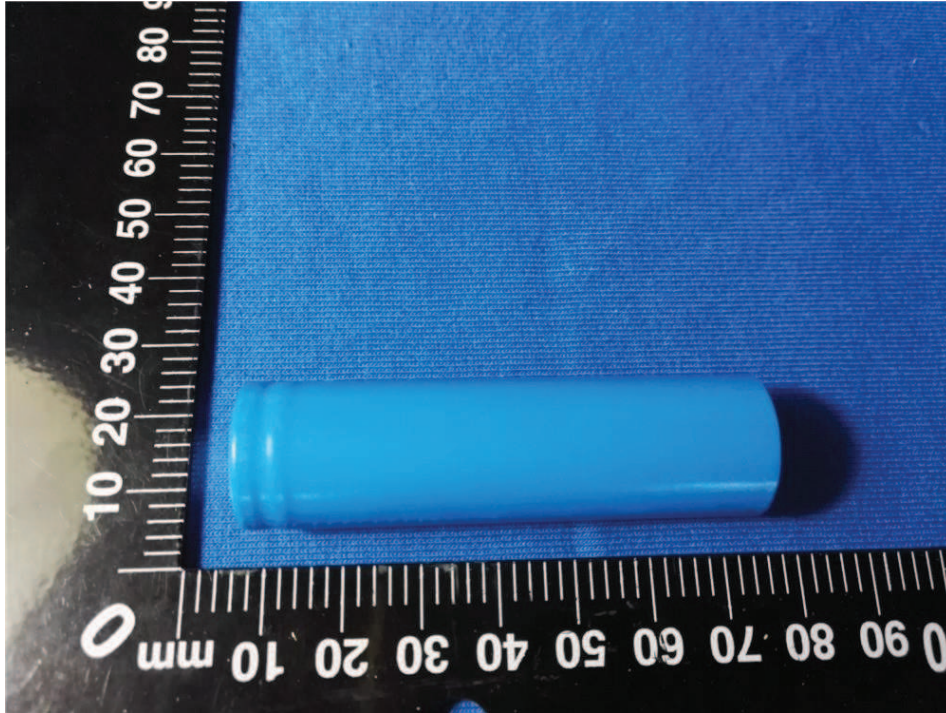


Figure 3 Front view of cell (model: IMR18650-1500mAh)

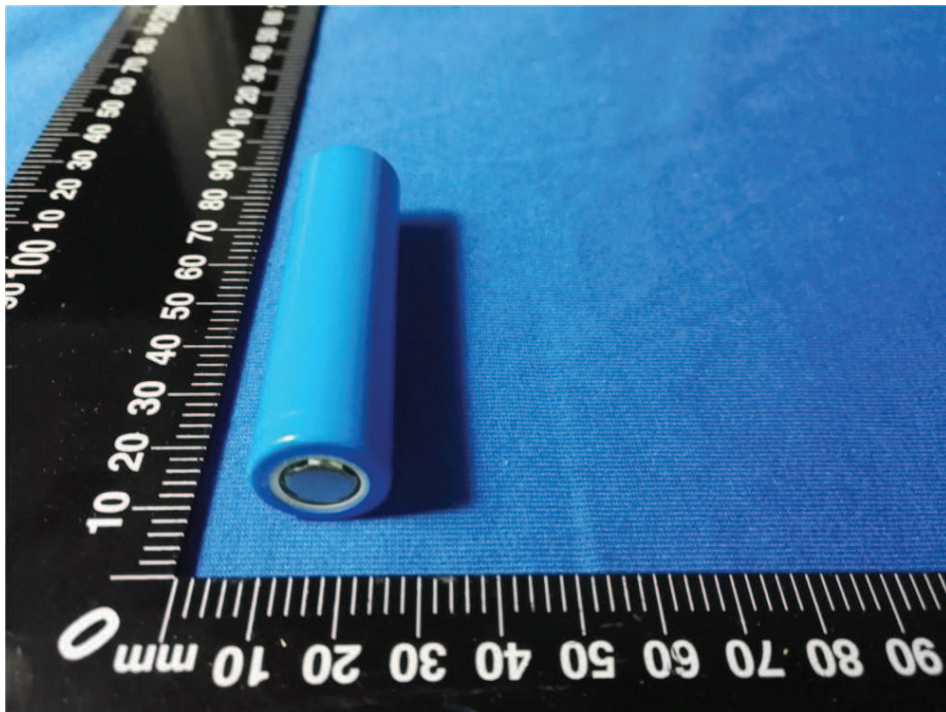


Figure 4 Back view of cell (model: IMR18650-1500mAh)